SynAT.Vis — How the Tool Works (Backend)

This document explains what happens when a transcript is submitted to SynAT.Vis, the algorithm behind each check, and the evidence that its output is trustworthy. The tool is a **deterministic, rule-based diagnostic**: the same input always yields the same output, every flag traces to a published mechanism, and it emits **no black-box score**.

1. What happens when you submit a transcript

Processing is entirely local — the sequence never leaves the machine. It flows through six stages:

1 UPLOAD
A transcript file (FASTA or GenBank / SnapGene) is opened locally.

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2 PARSE
Split into 5'UTR | CDS | 3'UTR; normalise (uppercase, U->T, validate ACGTN).

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3 LOAD HOST PROFILE
Cr-nuclear constants + data tables (codon usage, optimality, codon-pair, tAI, tRNA counts).

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4 RUN 9 DETECTION MODULES
codon | composition | cloning | polya | structure | uorf | instability | splice | silencing

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5 REMEDIATION
For each CDS motif flag, search the minimal synonymous edit that removes it (protein unchanged).

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6 RANK & REPORT
Flags grouped by module, ranked by severity; text + JSON. NO composite score.

2. The algorithms behind each check

Each module reads its constants from the host profile and applies one well-defined algorithm:

module	algorithm
`codon`	RCA = geometric mean of per-codon adaptiveness (Kazusa). Rare-codon cluster = >=5 codons with weight<0.10 in an 8-codon window. Plus the frontier panel: optimality (windowed log2 highly-expressed/genome), codon-pair score (Coleman), and tAI (dos Reis, from GtRNAdb tRNA gene counts).
`composition`	50-nt sliding windows (step 10); contiguous windows below 45% GC merged into troughs >=60 nt are flagged (low GC drives silencing in Cr). Homopolymer runs >=10.

`cloning`	Exact string match of Type IIS recognition sites (BsaI/BsmBI/BbsI/SapI) on both strands.
`polya`	Exact match of the Cr poly(A) signal TGTAA (+ variants) requiring a downstream 30-nt window of >=85% GC (the Cr near-site context).
`structure`	Fold the initiation window: ViennaRNA minimum free energy if installed, else a dependency-free Nussinov base-pairing maximiser. Flag if the paired fraction >0.70 (structured starts impede ribosome initiation).
`uorf`	Scan the 5'UTR for AUG...stop; classify by reading frame relative to the main start.
`instability`	Count AU-rich elements (ATTTA); flag a cluster of >=2 within 50 nt.
`splice`	Detect a Cr-consensus donor (GTRAG) + acceptor (YAG) forming an intron of 50-500 nt; GC-rich = intended intron (info), out-of-frame = accidental splice (low).
`silencing`	Heuristic and quarantined: re-reports GC troughs and prints strain/intron guidance; never a sequence verdict.
`remediation`	For a CDS motif, enumerate synonymous changes over the overlapping codons; keep the minimal set that removes the motif without recreating it, ranked by adaptiveness.

3. Why the response is believable

Trust rests on four independent things, none of which is 'trust the model':

3a. Every threshold is measured, not guessed

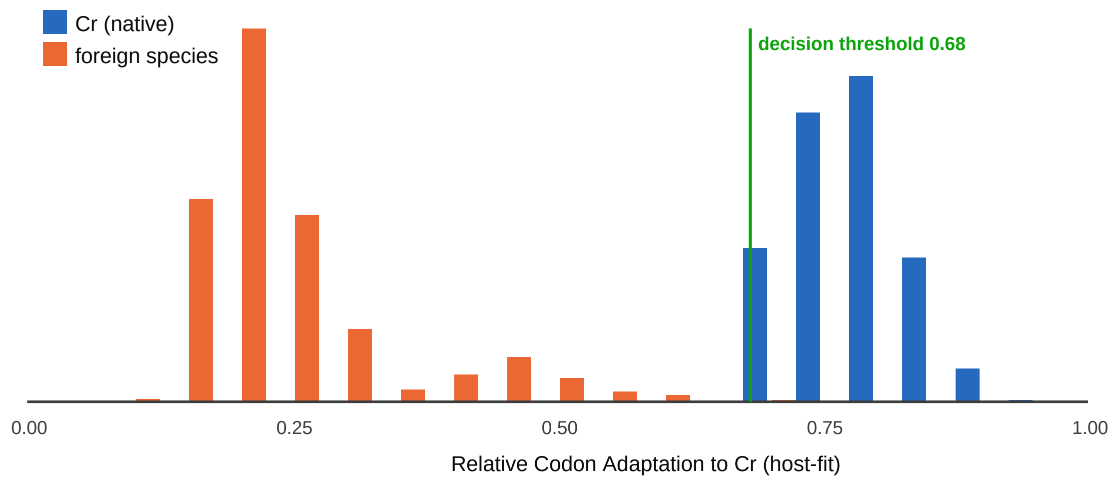
The tunable cutoffs were set by running the tool over **5,000 real Cr nuclear genes** and reading the false-positive rate directly (poly(A) ~2%, codon ~4%, composition ~1-2%, structure 2%). They then held on **4,500 genes the thresholds were never tuned on** — the operating points generalise, not overfit (see THRESHOLDS.md).

3b. Validated four independent ways

leg	what it proves	result
Leg 1 — specificity	native genes scan clean	FP 1-4% across 5,000 genes; held-out generalises
Leg 2 — sensitivity	planted motifs are caught	3/3 detected; detection curves via leg2sweep
Leg 3 — literature	real rescue pairs behave	100% / 100% on 6 real cases (GFP, luciferase, ...)
Leg 4 — cross-species	host-specific, not flag-all	AUC 1.000; see below

3c. Cross-species proof — it separates Cr from other species

We ran 1000 real Cr transcripts against 1020 foreign transcripts (human, yeast, Arabidopsis) from NCBI RefSeq. The host-fit metric separates them almost perfectly — **AUC 1.000**. A decision threshold trained on one half, evaluated on the held-out half: **accuracy 97%, specificity 100%** (only 1 foreign transcript in 510 slipped through). Foreign genes are flagged 99-100% of the time; Cr genes pass ~95%. The distributions barely overlap:



3d. Deterministic and traceable

There is no randomness and no opaque model: each flag names the motif, its location, the literature basis, and (where possible) an exact synonymous fix. The same transcript always produces the same report, and every rule is sourced.

4. What it does NOT claim (so you can trust what it does)

The tool is transcript-level: it is silent on protein folding, stability, and proteolysis, and it does not predict expression level or output a score. It speaks only for the *Chlamydomonas reinhardtii* nucleus. A clean report means 'no sequence-visible red flags for this host', which is exactly — and only — what the evidence above supports.